Quiz — 1.3.2 Databases — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Section A (8 marks, 1 each)

1. **B** — Relational databases reduce redundancy and anomalies through normalisation and linked tables.
2. **B** — A foreign key is the primary key of another table used to create a link.
3. **B** — A composite key combines two or more attributes to uniquely identify a record.
4. **B** — A junction/linking table (with a composite key) resolves M:N.
5. **A** — Referential integrity requires foreign keys to match a valid existing primary key (or be null).
6. **B** — 2NF removes partial dependencies (non-key attributes must depend on the whole key).
7. **C** — Atomicity = all steps succeed or all are rolled back.
8. **B** — Locking a record stops concurrent edits causing a lost update.

Section B (12 marks)

9. **[2]** Any two for 1 mark each: flat file uses a single table vs multiple linked tables; flat file has high redundancy vs reduced redundancy; flat file prone to anomalies/inconsistency vs consistent; flat file simple vs scalable/complex queries.

10. **[2]** 1 mark: no repeating groups and all values are atomic (single-valued). 1 mark: the table has a (defined) primary key.

11. **[2]** 1 mark: remove transitive dependencies — where a non-key attribute depends on another non-key attribute. 1 mark: move that attribute to a separate table (so every non-key attribute depends only on the primary key).

12. **[3]** Sample SQL:

```
SELECT StudentID, Surname
FROM Student
WHERE Form = '12A'
ORDER BY Surname ASC;
```

1 mark: `SELECT StudentID, Surname FROM Student`; 1 mark: `WHERE Form = '12A'`; 1 mark: `ORDER BY Surname ASC` (ASC may be omitted as it is the default).

13. **[2]** 1 mark: concurrent transactions execute as if isolated — intermediate/uncommitted states are not visible to other transactions. 1 mark: prevents interference/inconsistent reads when transactions run at the same time.

14. **[1]** Benefit: prevents the lost update problem / keeps data consistent. Risk: can cause deadlock (or block/slow other users). (Both needed for the mark.)

Section C (5 marks)

15. (a) **[1]** `MemberID` in the `Loan` table is the foreign key; it links to `MemberID` (primary key) in the `Member` table.

(b) **[3]** Sample SQL:

```
SELECT Member.Name, Loan.BookTitle
FROM Member
JOIN Loan ON Member.MemberID = Loan.MemberID;
```

1 mark: `SELECT` of Name and BookTitle; 1 mark: `FROM`/`JOIN` of both tables; 1 mark: correct `ON Member.MemberID = Loan.MemberID`.

(c) **[1]** Referential integrity prevents a `Loan` record being created with a `MemberID` that does not exist in `Member`, so there can be no orphaned loan pointing to a non-existent member.

Total: 8 + 12 + 5 = 25 marks